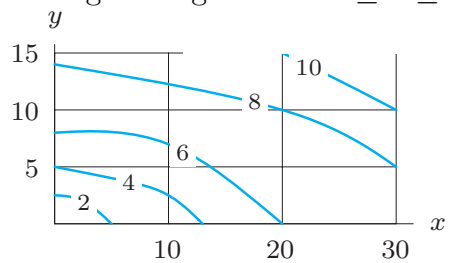
Suggested problems: 1-5 odd, 13, 15

Problem 1: The contour diagram of a function f is shown below. Estimate $\int_R f\ dA$ if R is the rectangular region with $0 \le x \le 30$ and $0 \le y \le 15$.



Take $\Delta A = 50$, and sample points in the upper left-hand corners of the subrectangles. Then

$$\int_R f\ dA \approx 50(4 + 5 + 7 + 6.6 + 7 + 8 + 8.5 + 9 + 10) = 3255.$$

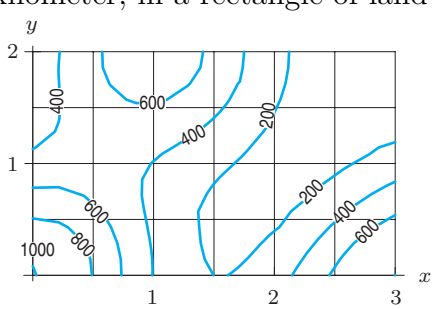
Problem 2: The table below gives the values of $f(x, y)$, the number of milligrams of mosquito larvae per square meter in a swamp. If x and y are in meters and R is the rectangle with $0 \le x \le 8$ and $0 \le y \le 6$, estimate $\int_R f\ dA$. Give units and interpret your answer.

y \ x	0	4	8
0	1	3	6
3	2	5	9
6	4	9	15

We take $\Delta A = 12$. Then with sample points in the upper right-hand corner, we get

$$\int_R f\ dA \approx 12(5 + 9 + 9 + 15) = 456 \text{ mg mosquito larvae.}$$

Problem 3: The figure below shows a contour plot of population density, people per square kilometer, in a rectangle of land 3 km by 2 km. Estimate the population in this region.



We take $\Delta A = 1$. If $p(x, y)$ denotes the population density function and R denotes the rectangle of land, then with sample points in the upper right-hand corner, we get

$$\int_R p\ dA \approx 1(400 + 100 + 300 + 700 + 250 + 100) = 1850.$$

Hence there are approximately 1,850 people in the rectangle of land.

Problem 4: If R is the rectangle $[0,2] \times [0,4]$, then evaluate

$$\int_R 2x \; dA.$$

From the graph of $z = 2x$, we get

$$\int_R 2x \; dA = 16.$$